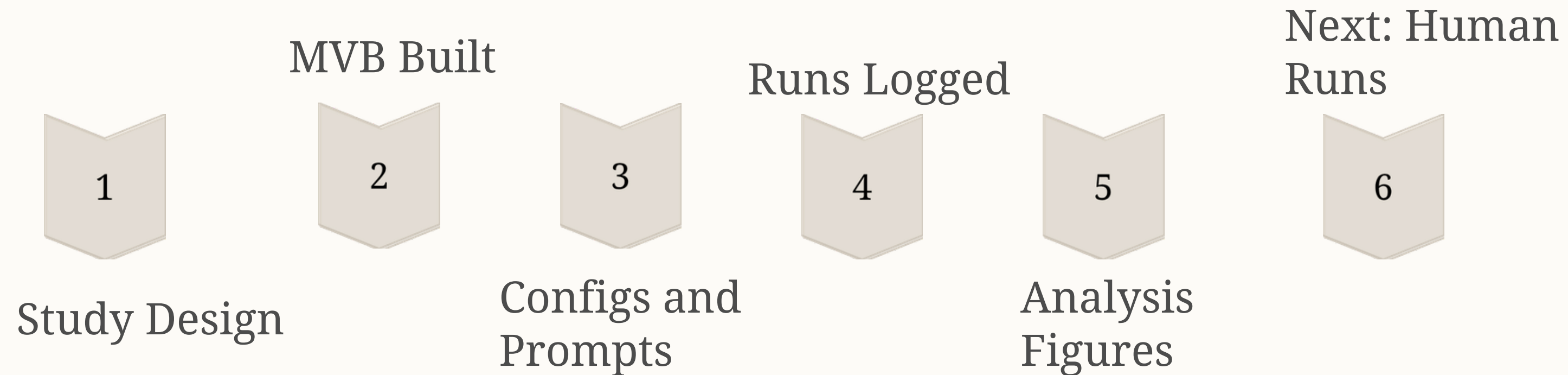


Sprint Update: From Design to Evidence

Conversational Clustering on UCDP GED_Event

Giacomo Vettore,
Jasnoor Singh,
Daniele Notarangelo



Minimum Viable Build Delivered

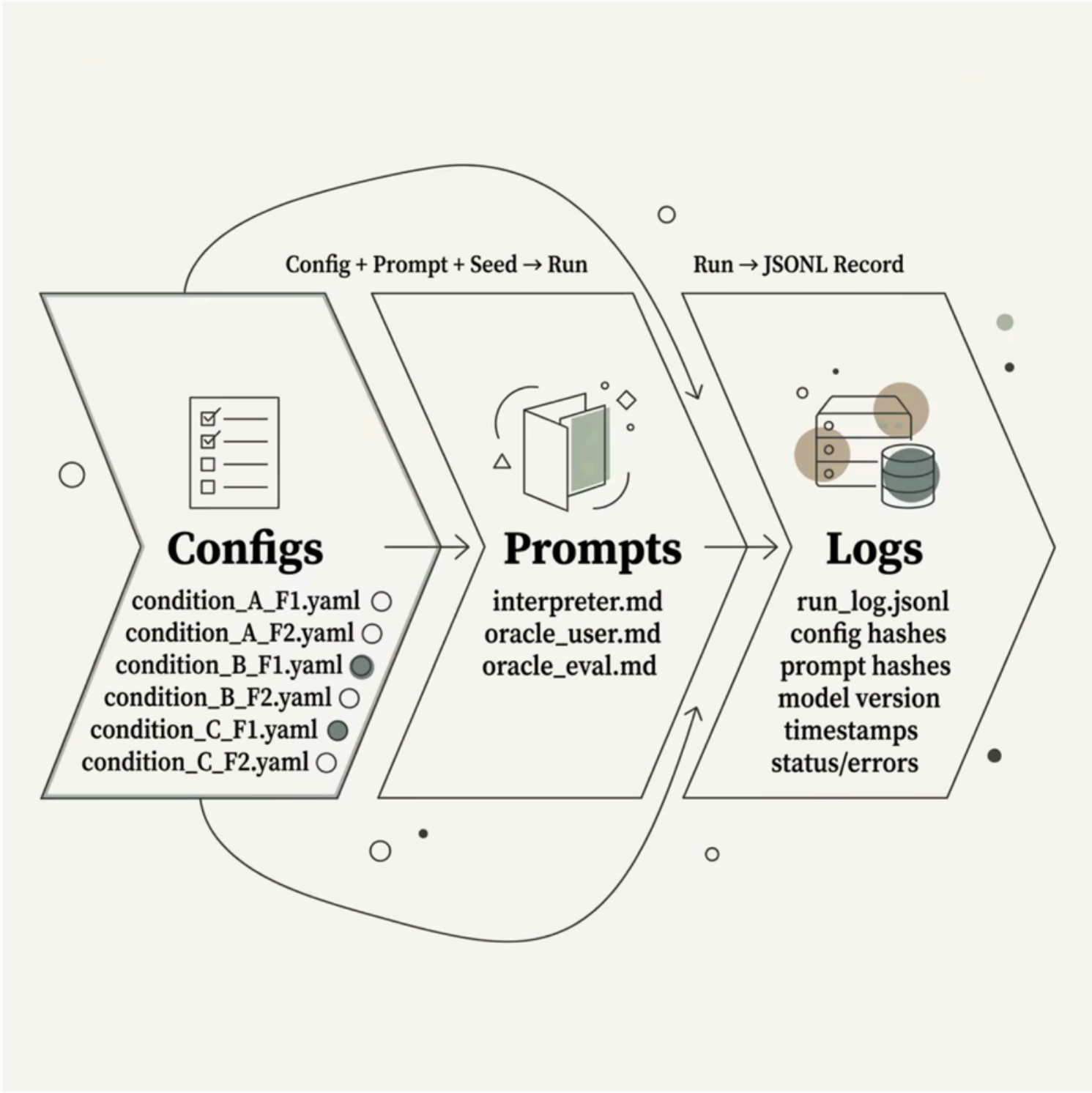
Implemented Modules

The build is the minimum executable apparatus required to run the study reproducibly.

- `src/features.py` — F1/F2 feature encoding
- `src/cluster.py` — k-means + metrics
- `src/loop.py` — conversational refinement loop
- `src/logger.py` — JSONL structured logging
- `src/llm.py` — Claude API wrapper
- `scripts/run_experiment.py` — main runner
- `scripts/check_integrity.py` — completeness check

Reproducibility Layer: Configs, Prompts, Logs

Every run is traceable: **config** → **prompt** → **model** → **output metrics**. Prompts, configs, and logs are versioned artifacts.



Audit Trail

Each experimental run produces a complete, versioned record linking configuration, prompt, model version, and output metrics.

K Finalized: Elbow Method

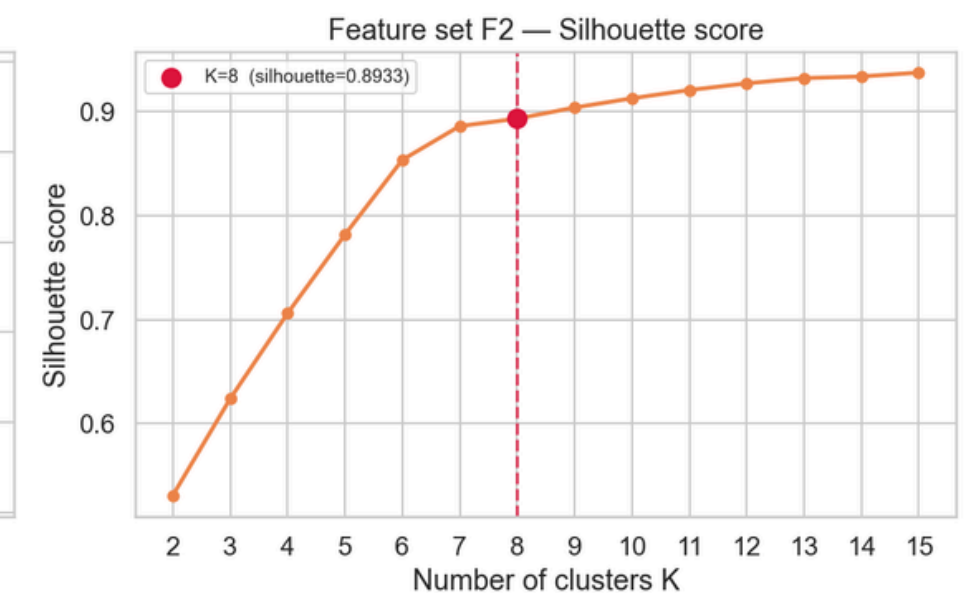
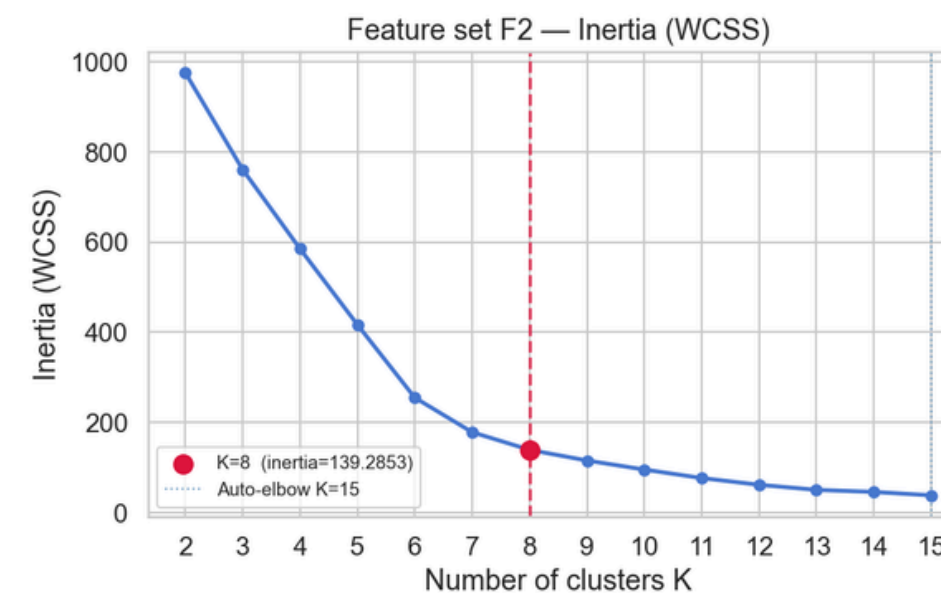
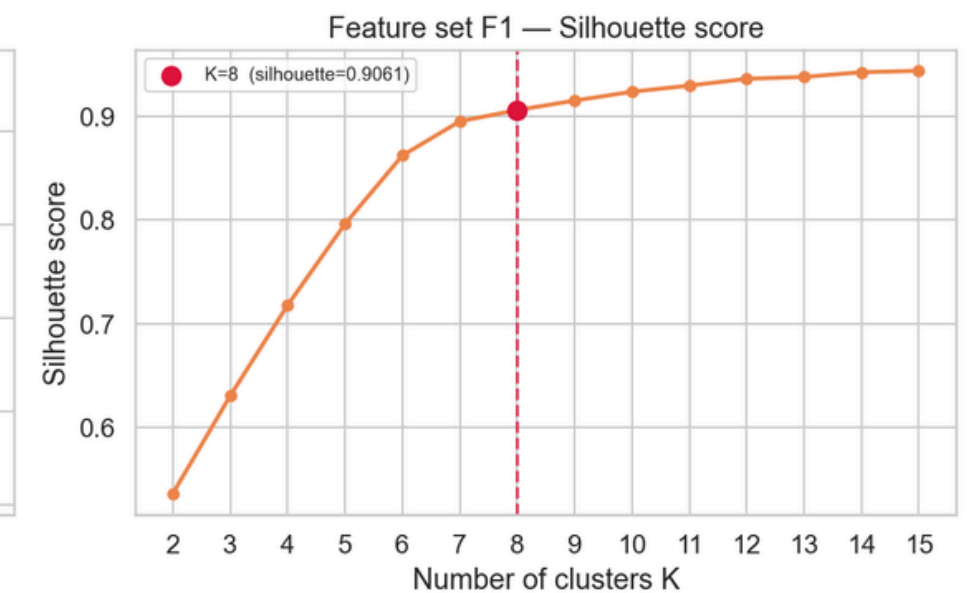
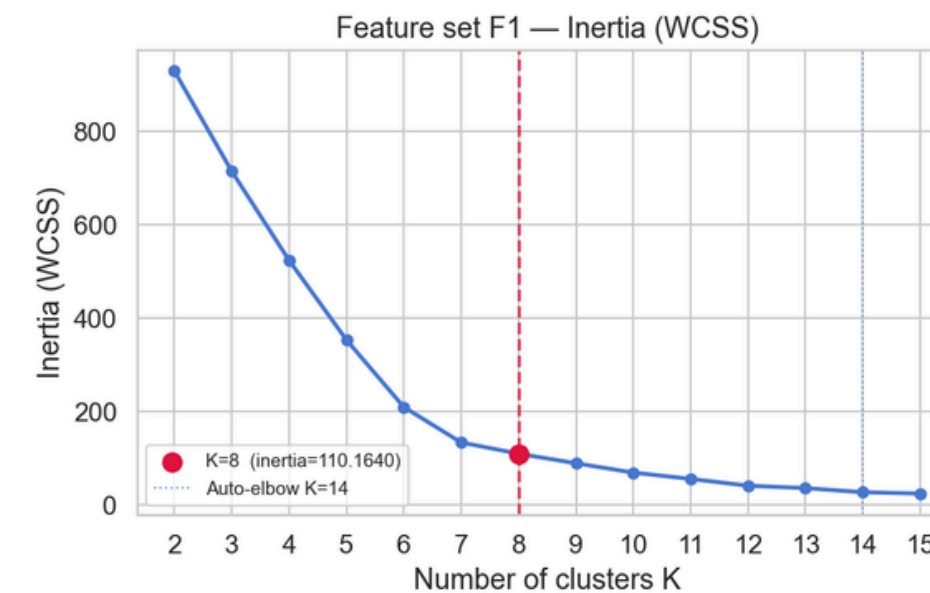
Methodological Decision

- K fixed at 8 for comparability across conditions A, B, C
- Elbow method applied before final runs
- Prevents conversational refinement from being confounded with K-selection



Scope limitation: the study does not test whether conversation helps choose K.

Elbow method — K = 2...15 on experimental sample (n=2,000, seed=42)
Vertical dashed line = K=8 used in the experiment



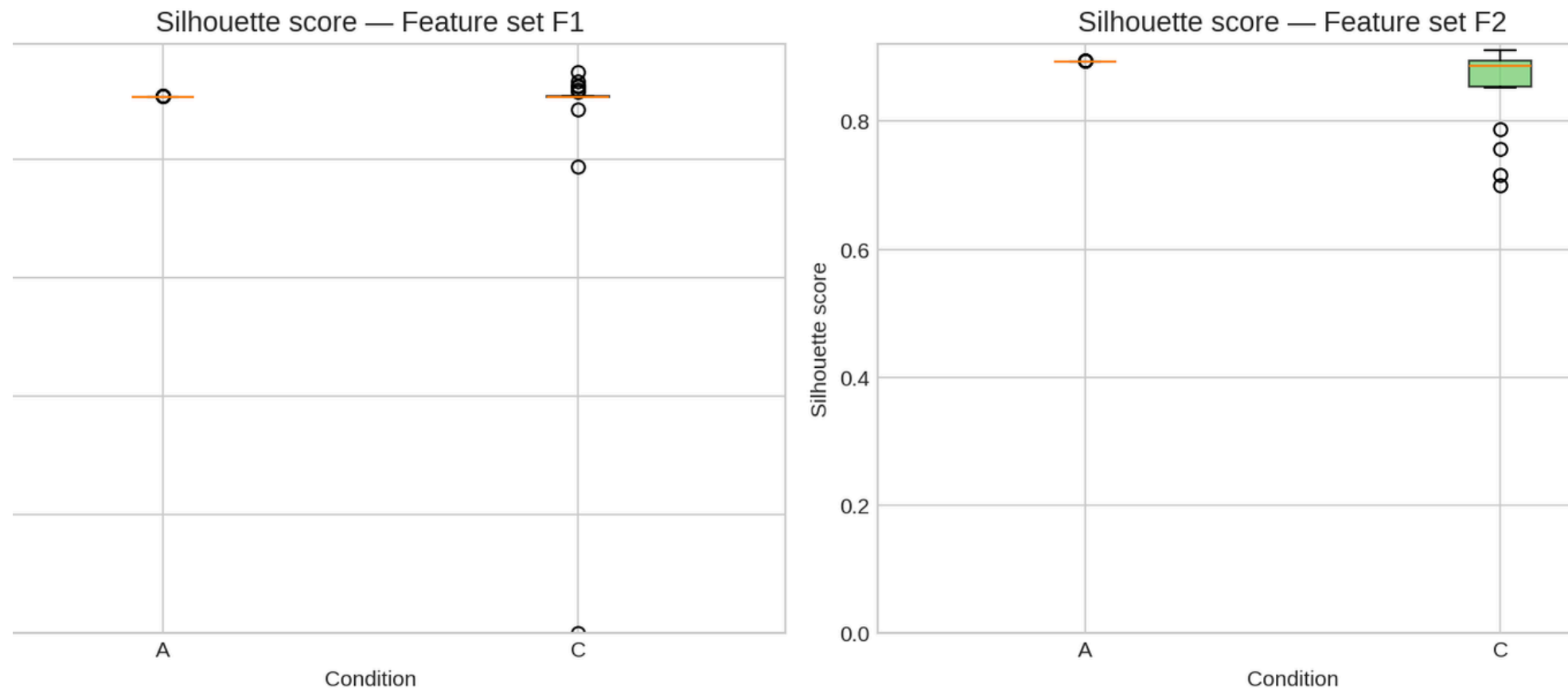
Execution Status: Runs Completed

Baseline (A) and LLM-oracle (C) conditions are complete. Human conversational runs (B) are the main remaining experimental gap.

Condition	F1 — Location-only	F2 — EDA-informed full
A — Baseline	30 / 30 ✓	30 / 30 ✓
B — Human Conversational	Pending	1 / 30
C — LLM Oracle	30 / 30 ✓	30 / 30 ✓

Preliminary Results: Baseline vs Oracle

Distribution of final Silhouette scores across 30 seeds



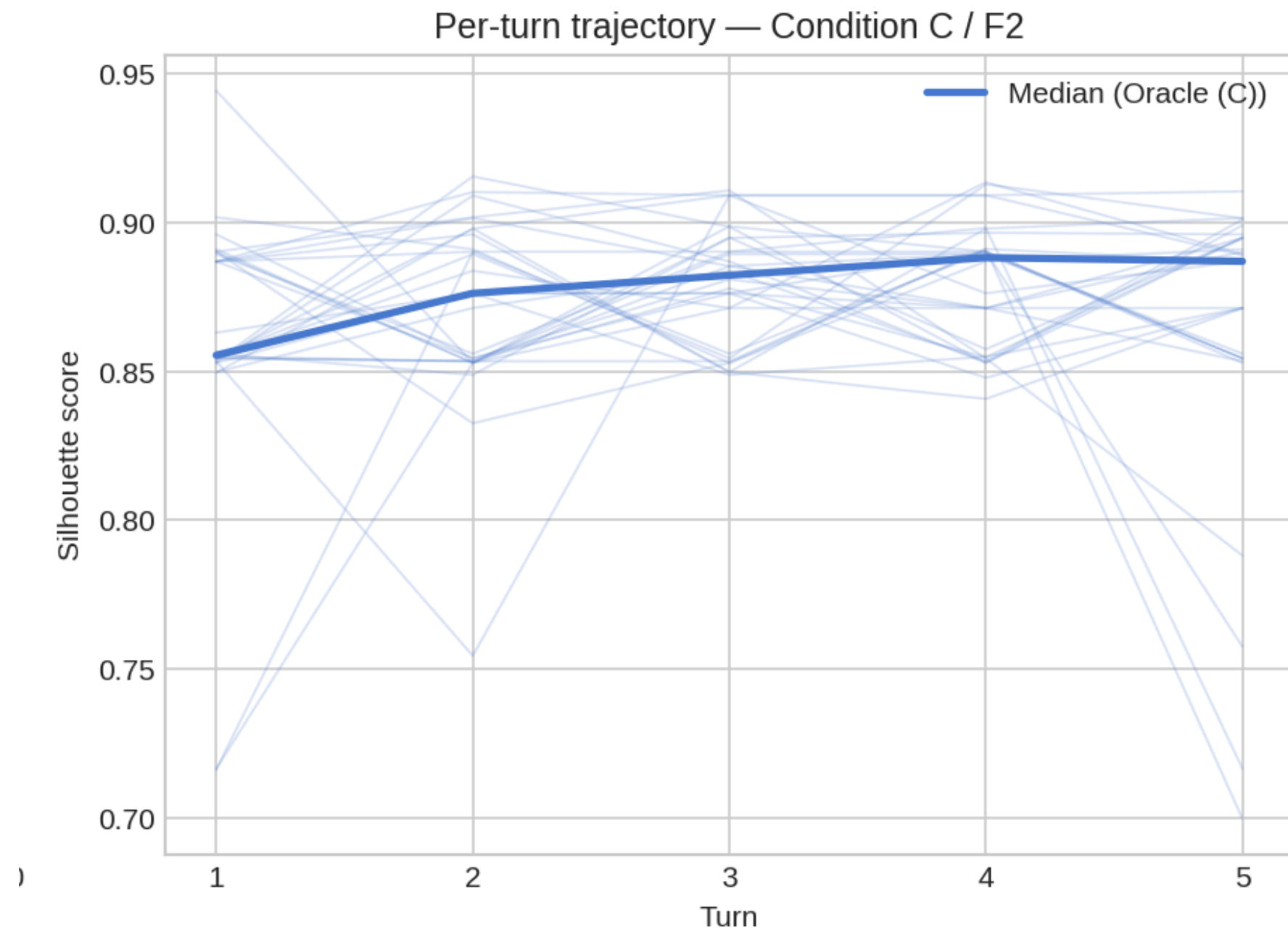
First Findings

- Baseline A performs strongly on Silhouette
- Oracle C does not clearly improve over baseline
- C/F2 vs A/F2: median improvement is **negative**
- Wilcoxon one-sided test does not support improvement



H1 is defined for human Condition B vs A
— oracle-only results do not yet settle the hypothesis.

Turn-by-Turn Refinement Trace



Conversational Loop Evidence

- Each oracle run logs up to **5 turns**
- Metrics tracked after each turn
- Notebook inspects whether refinement improves or destabilizes clustering
- Current evidence: trajectories are measurable, but improvement is not consistently stable

From final score to experimental trace.

What We Learned + Next Sprint

What We Learned

- MVB runs end-to-end
- Configs, prompts, and logs are versioned
- A and C conditions are reproducible
- Preliminary oracle-only results do not support improvement over baseline
- Project has moved from implementation to evidence

Next Sprint

- Complete Condition B human conversational runs
- Clean and standardize conversation records
- Run human rating protocol
- Compute H2 and H3
- Finalize confidence intervals and report figures
- Prepare final study viewer / presentation demo

The key next question is whether human conversational refinement succeeds where oracle-only refinement does not.